

ROSHITA JAISWAL

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EDUCATION

M.Sc. Data Science — Thakur College of Science & Commerce, Mumbai University

2025 – 2027

CGPA (Year 1): 8.91

B.Sc. Computer Science — Ratnam College, Mumbai University

2022 – 2025

Final Year CGPA: 9.55 | CGPA Trend: 8.00 → 9.40 → 9.55

WORK EXPERIENCE

Python & Robotics Trainer — Robomonnk

May 2025 – Present

- Design and deliver Python and robotics training sessions for student cohorts, breaking down core programming and robotics concepts into structured, hands-on lessons.
- Co-develop workshop curriculum and learning materials with fellow trainers, and guide students through live debugging to build independent troubleshooting skills.

PROJECTS

AI-Based Adaptive Traffic Signal Control System (Research)

- Built a computer-vision traffic-management prototype using YOLO for real-time object detection, analyzing traffic-density patterns to support emergency-vehicle prioritization and automated violation detection.
- Processed live video-stream data to identify traffic-flow trends and inform adaptive signal-timing logic in simulated scenarios.

Face Recognition Attendance System

- Developed a real-time attendance system in Python and OpenCV, applying data cleaning and preprocessing to live video input before detection, then structuring results for downstream analysis.

Gemini AI Assistant

- Built and deployed a Flask web application integrating Google's Gemini API to generate real-time AI-powered answers to user-submitted questions.
- Designed the end-to-end request flow from front-end form input through backend API calls to response rendering, with environment-based configuration for secure API authentication.

PUBLICATION

AI-Based Adaptive Traffic Signal Control System with Emergency Vehicle Prioritization and Automated Violation Detection

- Co-authored and published a research paper documenting the design, methodology, and real-time detection results of the traffic-signal system above.

TECHNICAL SKILLS

Languages: Python, SQL (MySQL), Java, JavaScript

AI & Machine Learning: Computer Vision (OpenCV, YOLO), Generative AI Integration (Google Gemini API), Machine Learning Fundamentals

Data & Analysis: Data Cleaning & Preprocessing, Exploratory Data Analysis, Pandas, NumPy, Excel (Pivot Tables, VLOOKUP/XLOOKUP)

Tools & Platforms: Flask, Git/GitHub, VS Code, XAMPP, Relational Database Design

ACHIEVEMENTS

- Published research analyzing AI-based adaptive traffic-signal systems, covering methodology, detection results, and system design.
- Maintain public GitHub repositories spanning computer vision, generative AI integration, and web application development.